Analysis of Factors Influencing Fire Incident Financial Loss in Toronto

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GitHub Repository

The complete reproducible pipeline, including data acquisition scripts, is available here:
https://github.com/JYW5114/JSC370-project

Introduction

This project investigates the factors influencing Fire Incident Financial Loss within the City of Toronto. This metric is defined as the estimated dollar loss associated with a fire incident, including property damage and related losses. Understanding the drivers of financial loss is critical for optimizing urban resource allocation, improving fire prevention strategies, and minimizing economic impact.

Research Questions: What factors influence the financial loss of fire incidents? How do weather conditions, property characteristics, and response resources relate to the severity of the loss? Are certain spatial and structural features consistently associated with higher losses?

Research Hypothesis: The model predicts financial loss immediately after a fire is brought under control, before the official damage assessment is completed. We hypothesize that incident characteristics, property type, response resources, and weather conditions contain enough information to approximate the final financial loss. If successful, such predictions could provide an early estimate of economic impact and help prioritize post-incident response and resource allocation.

Methods

Data Acquisition

To satisfy the course requirement for dynamic data retrieval, I integrated two primary **API** sources:

1. **Toronto Open Data API:** I retrieved the `fire-incidents` dataset by the CKAN API using the `requests` library. This dataset provides comprehensive logs of every fire emergency call from 2011-01-01 to 2024-12-31. Each record corresponds to a single incident and includes information about incident location, property characteristics, response resources, and fire suppression timelines.

2. **Open-Meteo Historical Weather API:** To examine the environmental context of fire incidents, I utilized the Open-Meteo Historical Weather API. I queried hourly data for temperature (2m), snowfall, and precipitation from January 2011 to December 2024, using the specific coordinates for Toronto (Lat: 43.65, Lon: -79.38).

Data Cleaning & Wrangling

- **Preparation:** I used `dt.floor('H')` on the fire dataset to enable a precise `left join` with the weather API output. I filtered the dataset to include only incidents with non-missing `Estimated_Dollar_Loss` values and removed extreme outliers using a log-transform approach to stabilize variance. Variables like `Incident_Station_Area` contained a mix of integer and string types, so data type unification was performed. Because the dataset includes incidents other than fires (e.g., medical calls), the data was further filtered to retain fire-related incidents only.

Detailed code of this process and data acquisition from the APIs can be found here: **https://github.com/JYW5114/JSC370-project/blob/main/dataset.ipynb**

The prepared dataset contains 36564 rows(observations) and 51 columns (features).

- **Feature Engineering:** I constructed a comprehensive set of temporal, spatial, operational, and risk-related features to enhance predictive performance and interpretability.
  - **Temporal features:** From the event timestamp (`Hourly_Timestamp`), I derived `Hour_of_Day`, `Month`, and `is_weekend` to capture diurnal, seasonal, and weekly patterns. To better represent cyclic time structure, I also encoded hour using sinusoidal transformations (`Hour_sin`, `Hour_cos`) to preserve periodicity.
  - **Response dynamics:** I defined `Response_Time` as the elapsed time between TFS alarm and arrival, and `Control_Duration` as the time from TFS arrival to the moment the fire is declared under control. These variables capture emergency response efficiency and fire suppression duration.
  - **Spatial features:** I computed `distance_to_downtown` using the Haversine formula to measure geographic distance from Toronto's downtown, enabling analysis of spatial heterogeneity in fire incidents.
  - **Incident and structural categorization:** I engineered domain-specific categorical features including `Property_Group` (e.g., Residential, Commercial), `Scale` (based on number of responding apparatus), `Area_of_Origin_Group`, `Fire_Extent_Grouped`, `Smoke_Spread_Binary`, and `Level_Of_Origin_Group`. These transformations reduce high-cardinality fire descriptors into interpretable, risk-relevant categories.
  - **Safety and system indicators:** I created binary and standardized indicators such as `Alarm_Binary` (fire alarm presence), and `Sprinkler_System_Presence`to capture building safety infrastructure and fire spread conditions.
  - **Risk-based features:** I derived `Ignition_Risk_Group` by ranking ignition sources based on median estimated dollar loss and discretizing them into risk tiers. I also included `Fire_Severity_Arrival` to represent the initial severity of the fire upon TFS arrival.

- **Preventing Data Leakage:** The full dataset was first split into training, validation, and testing sets. Exploratory analysis and model development were conducted only on the training data. In accordance with our research hypothesis, variables that become available after the fire is brought under control were excluded from the model, such as `Count_of_Persons_Rescued`, `Estimated_Number_Of_Persons_Displaced` and `Business_Impact`.

Statistical Tools

Exploratory data analysis (EDA) was conducted using Python libraries including pandas, seaborn, matplotlib, and plotly to examine the distribution of fire-related financial loss and explore potential relationships between predictors and the target variable. Visualizations such as histograms, boxplots, line plots, geographic maps, and scatter plots were used to investigate distributional patterns, temporal trends, and spatial heterogeneity across Toronto. In addition, hypothesis testing was applied to assess whether fire safety systems (e.g., alarms and sprinklers) are associated with statistically significant differences in financial loss.

Generalized Additive Models (GAMs) were used to capture potential non-linear relationships between key continuous predictors (e.g., temperature and time of day) and log-transformed financial loss.

Modeling Approach The primary predictive task was addressed using Random Forest and XGBoost models, trained on log-transformed financial loss to reduce skewness and improve stability. Both models incorporated incident characteristics, weather conditions, response resources, and other engineered features. Hyperparameters were optimized using cross-validation with RMSE as the primary evaluation metric.

Among the best models of each family, XGBoost outperformed Random Forest in validation and was selected as the final model. It achieved lower RMSE and higher R^2 in both log and dollar scales.

Diagnostic analysis revealed that prediction errors were larger at the extremes of the loss distribution, as reflected in the gap between MAE and RMSE. Residual plot Figure 2 showed mild skewness, indicating systematic overestimation for certain incidents. This pattern could also be observed in the predicted vs actual scatter plot Figure 1, with overestimated zero-loss data points.

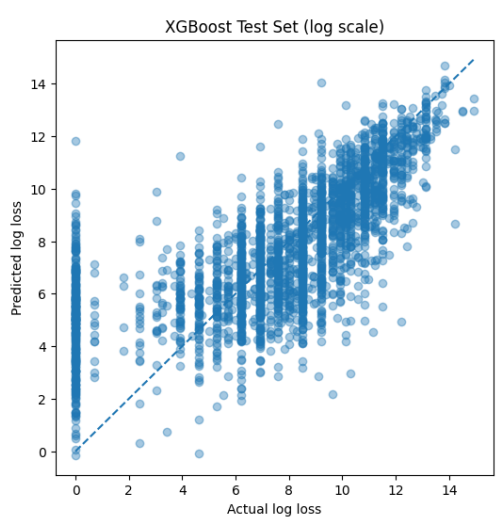

Figure 1: Predicted Loss vs Actual Loss by XGBoost

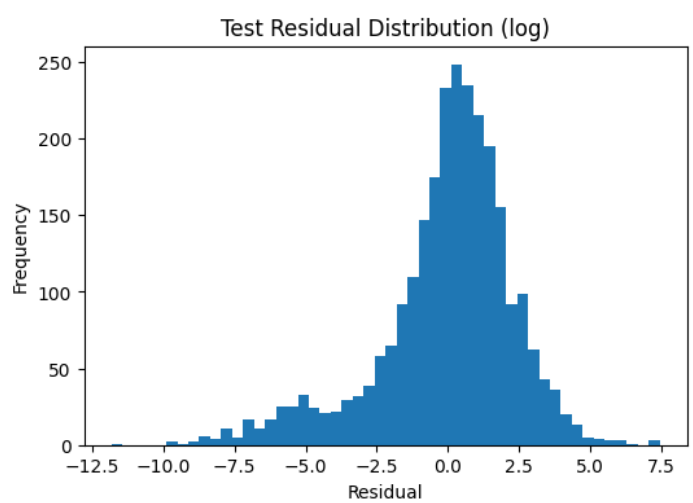

Figure 2: Distribution of Residuals (actual minus predicted log loss)

To further improve performance, a two-stage modeling strategy separating zero- and non-zero-loss incidents was explored but did not yield improvements because the classifier misclassifies many true zero-loss cases as non-zero, forcing the regression model to assign them positive predictions. This likely occurs because many zero-loss cases are not truly zero but reflect reporting issues or missing values, making the classification boundary inherently noisy.

Overall, the single-stage XGBoost model with log-transformed targets provided the most stable and reliable performance, capturing substantial signal from engineered features while remaining limited in zero-loss cases.

Results

1. Spatial distribution of fire in Toronto by Year

It's the first interactive visualization on my website. **To My Website**

Fire incidents in Toronto are generally spread across the city, although the downtown area exhibits the highest density of incidents. Most fires appear to be controlled relatively quickly. However, incidents associated with larger financial losses and longer control durations are more frequently observed in the downtown and northwestern areas.

The plot suggests that the relationship between fire duration and financial loss is not straightforward. Some incidents with relatively short control durations still result in substantial financial damage, while fires that take longer to control do not necessarily lead to extremely large losses.

2. Distribution of the financial loss

The overall distribution of financial loss is highly right-skewed, as shown in Figure 3. Most incidents result in relatively small losses, while a small number of fires lead to extremely large financial damages.

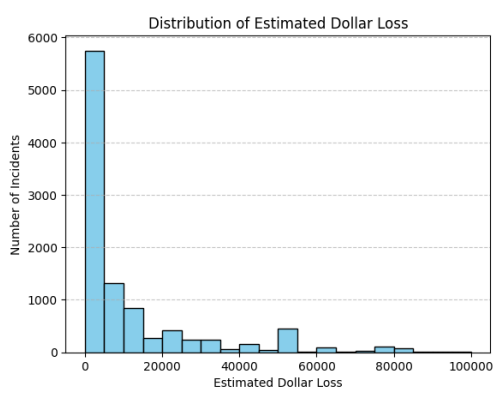

Figure 3: Distribution of Estimated Dollar Loss

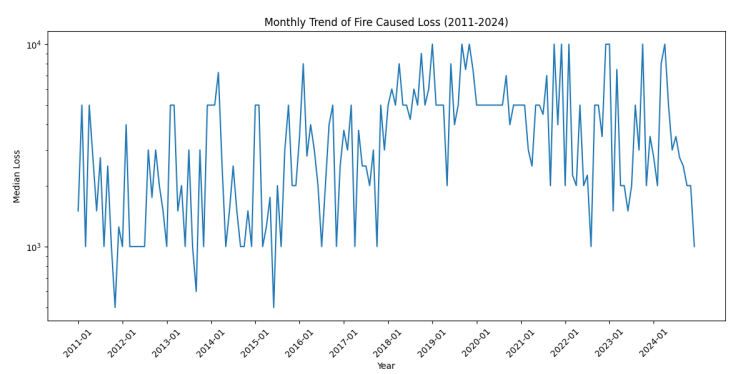

Figure 4: Monthly Trend of Fire Caused Loss

To further summarize the key variables used in the analysis, Table 1 presents several descriptive statistics including the mean, median, standard deviation, and range.

Table 1: Summary statistics for key variables in the fire incident dataset

	mean	median	std	min	max
Estimated_Dollar_Loss	38305.75	3000.0	209327.10	0.0	13000000.00
Control_Duration	833.53	404.0	1337.06	1.0	14379.00
temperature_2m	9.99	9.8	10.49	-24.5	33.30
snowfall	0.01	0.0	0.09	0.0	3.36

Examining the monthly trend of fire-related losses in Figure 4 reveals a recurring pattern across many years: a noticeable downward spike appears around the middle of the year, where total losses temporarily decrease before rising again. This suggests the presence of a potential seasonal or periodic effect. Overall, the total financial losses remain relatively stable throughout the study period, with a noticeable increase between 2018 and 2021, followed by a slight decrease from 2022 to 2024.

3. Temporal Features

By analyzing the relationship between month, temperature, and financial loss in Figure 5, a clear periodic pattern can be observed. While temperature follows a strong seasonal cycle across months, it also captures underlying variation more directly, as substantial temperature differences can exist within the same month.

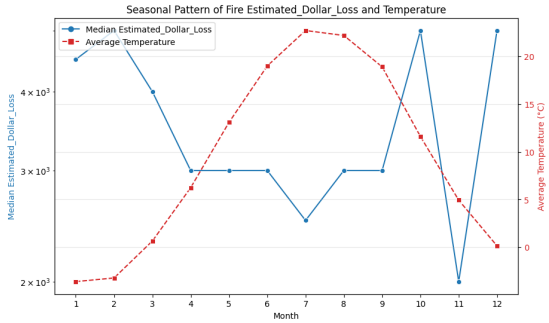

Figure 5: Seasonal Pattern of Fire Loss and Temperature

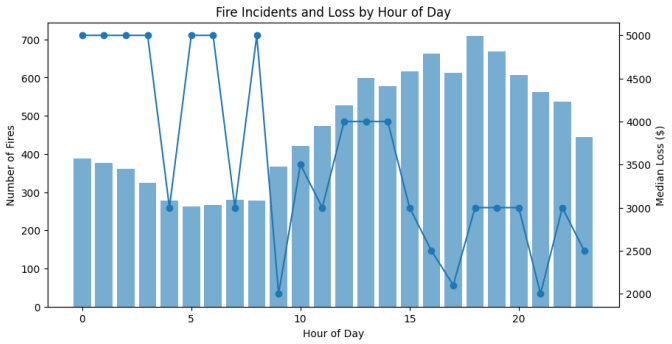

Figure 6: Daily Pattern of Fire Loss

Figure 6 reveals diurnal variations in fire-related dollar losses, with incidents occurring during the night showing a lower frequency but a higher median loss. To represent this cyclical nature effectively, I encoded the 'Hour of Day' as a periodic variable using sine and cosine transformations in the model.

4. Response Time across Wards and Loss Levels

It's the second interactive visualization on my website. **To My Website**

Overall, median response times are similar across loss levels in most wards. However, Wards 24, 25, 26, and 39 show consistently slower responses, while Wards 13, 27, 28, and 29 are generally faster.

Some wards (15, 16, 22, 33, 36) exhibit longer response times specifically for high-loss incidents, suggesting that delays in these areas may contribute more directly to severe outcomes. In contrast, wards such as 4 and 13 maintain relatively fast response times even in high-loss cases, indicating

that factors like fire severity at arrival or asset vulnerability may play a larger role than response delay.

Response times are also more variable in Wards 1–15 compared to 26–44, reflecting greater operational variability. Additionally, many extreme high-loss outliers are associated with vehicle fires, suggesting higher volatility and rapid escalation in these incidents.

5. Weather Effects

The weather also appears to influence the loss. Figure 7 shows a slight increase in financial loss when snowfall reaches moderate or higher levels. Snow may slow response times or create additional operational challenges, potentially allowing fires to spread before they are fully controlled. However, the precipitation doesn't show the same pattern, with different levels of rainfall generally associated with somewhat similar losses.

6. Fire Suppression Resources and Property Uses

This Dashboard presents a parallel categories (Sankey) plot, which is used to examine how `Property_Group`, `Sprinkler_System_Presence` and `Scale` jointly interact and influence fire loss outcomes. Here, incident scale is defined by the number of responding apparatus: Small (≤ 5) and Large (> 5). Each path in the diagram represents a combination of these attributes, while the color encoding reflects the Loss_Level (Low, Medium, High by 0.33 and 0.66 quantiles).

Open Dashboard

The underlying risk of a specific fire scale is approximately similarly distributed across property groups. In general, larger-scale incidents are more likely to be associated with higher financial losses. Properties equipped with sprinkler systems tend to exhibit a relatively lower risk of generating large losses.

In Residential and Commercial groups, as fire suppression improves from No to Partial and to Full sprinkler coverage, the proportion of high-loss (dark purple) outcomes decreases in cases of large scale. This suggests that sprinkler systems play a strong mitigating role in reducing severe financial loss when incidents escalate. However, this relationship is not equally evident in the Other property category.

7. Other Effects

Besides all the findings mentioned above, I also examined other categorical features including `Area_of_Origin_Group`, `Smoke_Spread_Binary`,`Level_Of_Origin_Group`, `Ignition_Risk_Group`, and `Alarm_Binary`, all of which shows different loss distribution across groups with a confirmation from hypothesis testing.

8. Correlation Matrix

The correlation matrix in Figure 8 shows that control duration and the number of responding apparatus have the strongest positive relationship with financial loss, while weather variables show very weak correlations. This does not preclude the importance of weather condition. As previously explored in the snowfall analysis (Figure 7), environmental impact manifests through non-linear threshold effects rather than direct linear scaling. Therefore, these features should be retained for

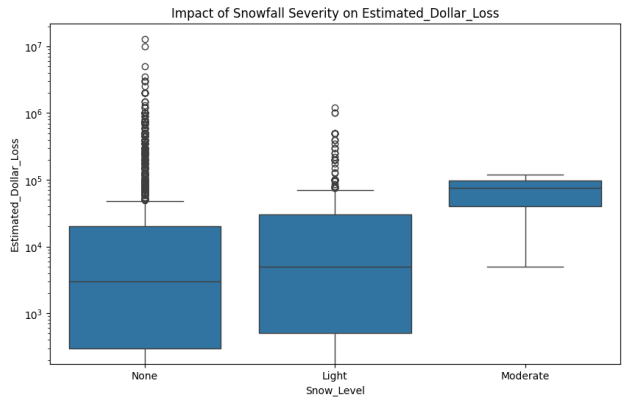

Figure 7: Impact of Snowfall on Loss

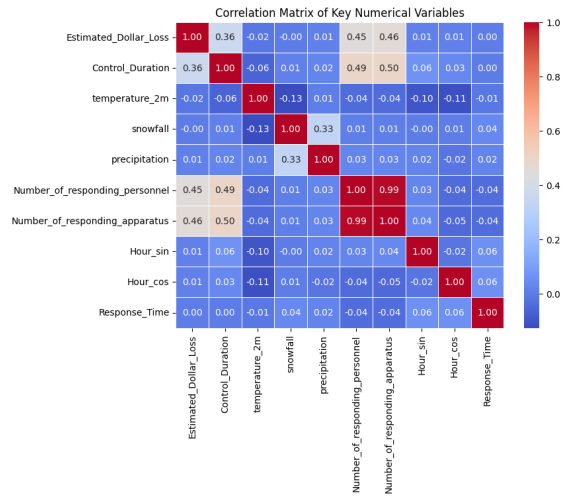

Figure 8: Correlation Matrix of Numerical Variables

machine learning models capable of capturing complex interactions, such as those between freezing temperatures and operational response challenges.

9. Non-linear Weather and Temporal Effects (GAM)

While linear correlations for weather variables appear weak, Generalized Additive Models (GAMs) reveal strong non-linear effects once other factors are controlled for, as shown in Figure 9. The results suggest a pronounced diurnal pattern, with higher financial losses occurring during early morning hours, consistent with delayed fire detection at night. Temperature exhibits a clear threshold effect, where extreme cold (below -20) is associated with substantially higher losses, while moderate temperatures show relatively stable impacts. Snowfall also shows a generally increasing relationship with loss, although uncertainty grows at higher levels due to data sparsity, indicating that severe snow events are less frequent but potentially more damaging.

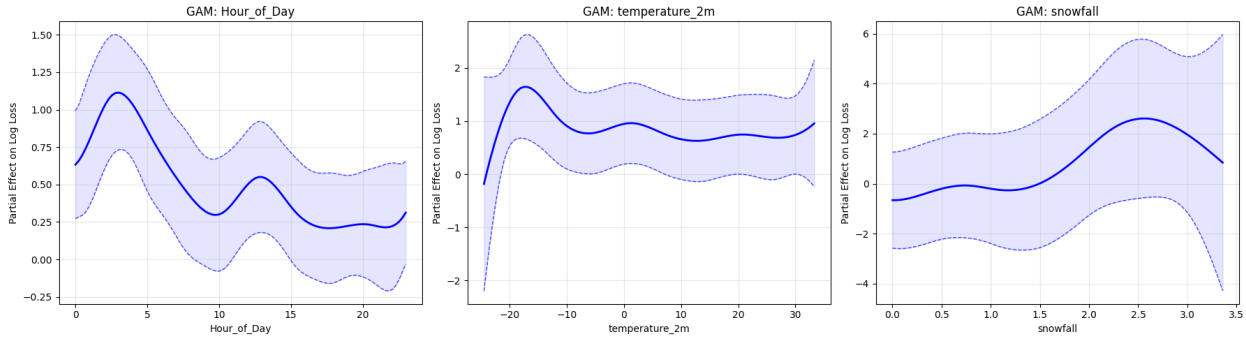

Figure 9: GAM Partial Dependence Plots for Temporal and Weather Features

Modeling

After model selection, XGBoost with tuned hyperparameters (learning rate = 0.05, max depth = 4, subsample = 0.7, and 400 estimators) outperformed other models. Validation performance is

Table 2: Summary Statistics for the Best Models in the Family

Model	CV RMSE (log)	Val RMSE (log)	Val RMSE ($)	Val R2 (log)	Val R2 ($)
Random Forest	2.493	2.414	125686.481	0.515	0.267
XGBoost	2.474	2.363	120589.685	0.535	0.326

Table 3: Test Set Performance of the XGBoost Model

RMSE (log)	MAE (log)	R2 (log)	RMSE ($)	MAE ($)	R2 ($)
2.410	1.747	0.532	119814.139	30681.905	0.377

summarized in Table 2.

The most important features of the XGBoost model are presented in Figure 10. The features reflecting the scale of the fire are the most important, including the extent of the fire and smoke, as well as the number of responding apparatuses. Other features such as property type, ignition source, and the safe system availability are also considered important. Although weather conditions are not on the top list, they also play an important role as stated in the previous sections.

The predicted versus actual losses on the test set (log scale) are shown in Figure 1, illustrating instability around zero. The detailed test statistics are summarized in Table 3.

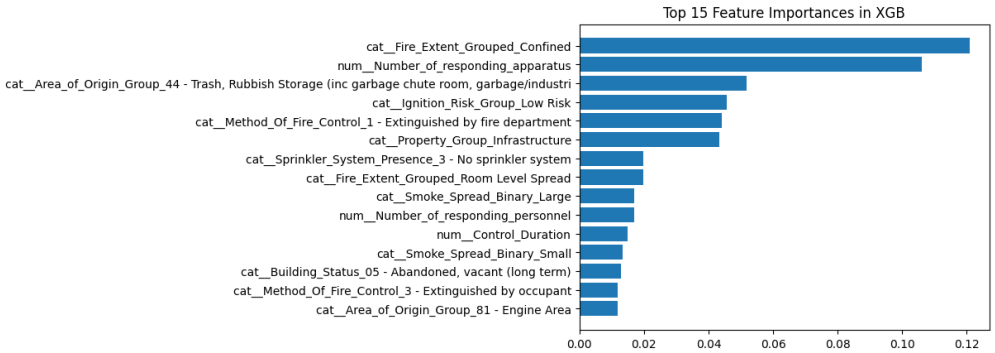

Figure 10: Feature Importance of the Selected Model

Conclusions and Summary

This study shows that fire-related financial loss in Toronto is driven by a combination of response efficiency, property characteristics, and fire severity.

The XGBoost model achieves moderate predictive performance, indicating that although key patterns are captured, substantial uncertainty remains – particularly for zero-loss incidents. This reflects limitations in both data noise and unobserved factors. Future improvement may include enhancing data quality and upgrading modeling techniques.

From a practical perspective, the results suggest that improving early detection systems like alarms, optimizing response in slower wards, and preparing for extreme weather conditions could help reduce financial losses. Overall, the model provides useful early estimates, but should be used as a decision-support tool rather than a precise predictor.